Standardizing Flight Software Architecture for Interplanetary Small Satellites

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Outline

- **Motivation**
- **Benefits**
- **A Path To Collaboration?**
- **Case Study of JHU/APL's Adoption of cFE**
  - Overview of cFE
  - Implementation of cFE on the Van Allen Probes
  - Lessons Learned
- **Roadmap**
- **Questions**

Motivation

- **Flight software is critical to many subsystems on a spacecraft.**
- **With faster processors on the horizon, flight software functionality will continue to rise.**
- **Given the mass and power limitations for interplanetary small satellites, flight software plays a critical role in the downlink bandwidth and instrument capabilities.**
  - Software can help with these constraints, but with more software comes increased costs and system complexity.
- **Small spacecraft also have smaller budgets—repeated development of software has a negative impact**
- **We propose that the community look towards a common software flight software architecture.**

Benefits

- **Reduced Cost**
- **Reusability across agencies**
- **Allow developers to focus on new features needed for a particular spacecraft**
  - No need to re-write the same common applications
- **Portability across platforms**
- **Identification of key core applications are necessary to reduce costs**

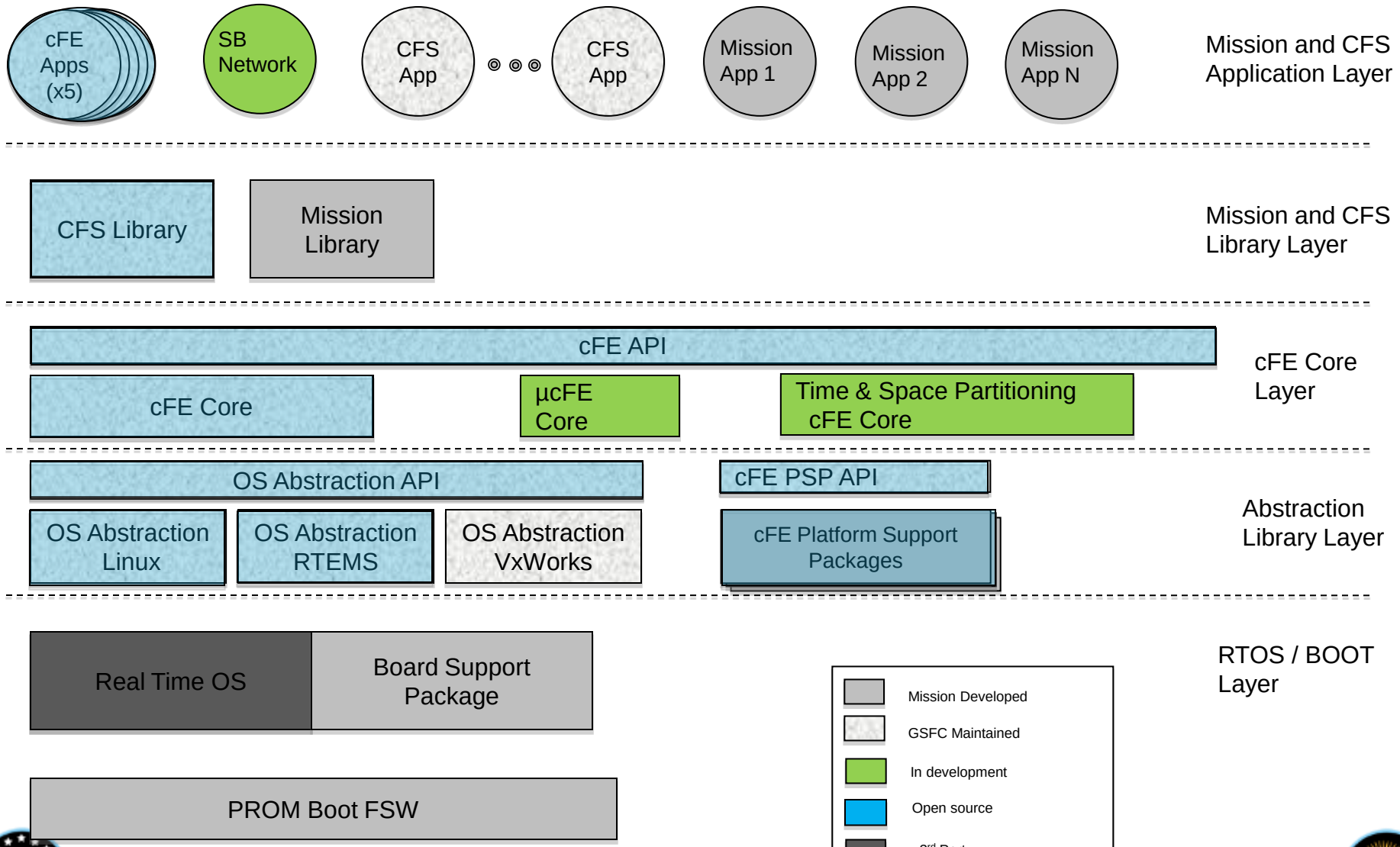
A path to collaboration?

- **Commercial products**
  - Costly for Universities or small companies
- **Home-grown solutions**
  - Forces long-term support to occur locally
  - Heavy and costly initial investment
- **Open source**
  - Core Flight Executive (cFE)
  - Developed by NASA Goddard Space Flight Center
  - Used by multiple NASA Centers

Core Flight Executive

- **The Core Flight Executive (cFE) was developed at NASA Goddard Space Flight Center.**
  - Portable, platform independent architecture for developing Flight Software "Apps"
  - Written in C and dependent on the Operating System Abstraction Layer (OSAL), also developed at GSFC.
- **JHU/APL has adapted cFE to include common mission independent applications.**
  - Used on the Van Allen Storm Probes, Robotic Lunar Lander (RLL), and Solar Probe Plus
- **Also used by multiple NASA centers**
  - ARC (LADEE), JSC (Morpheus), MSFC (RLL), GRC

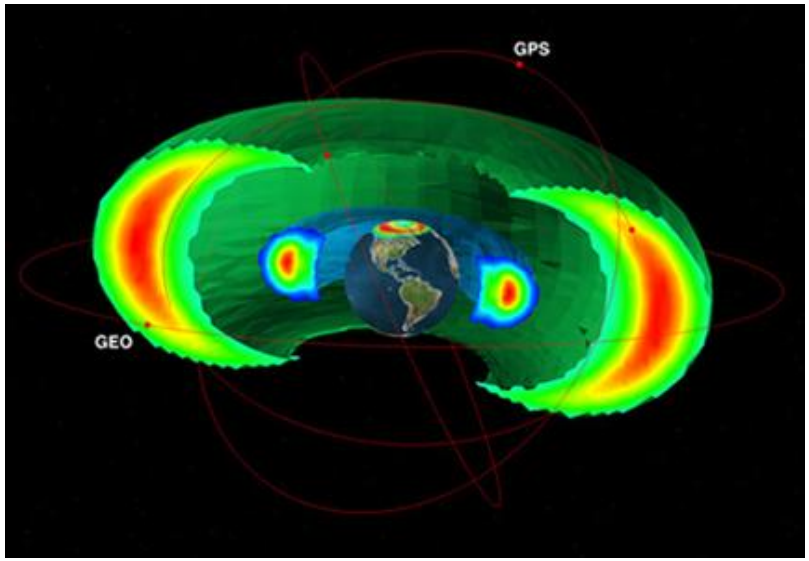
CFS Software Layers and Components



NASA's Van Allen Probes

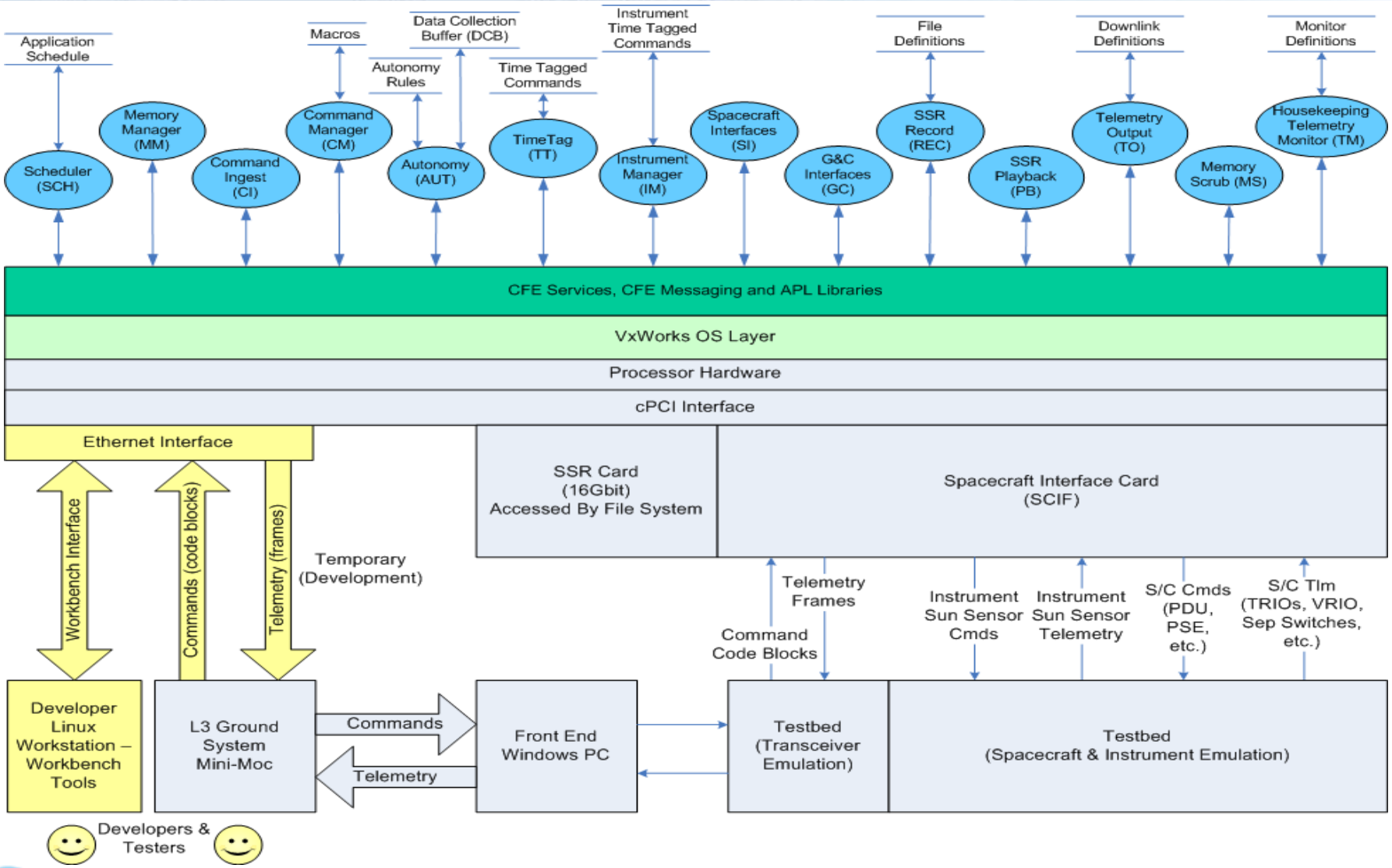
- **Launched August 30, 2012**
- **Two spacecraft studying the effects of the Van Allen Belts**
- **First use of cFE by JHU/APL**

http://www.nasa.gov/mission_pages/rbsp/mission/index.html



http://www.nasa.gov/mission_pages/rbsp/science/index.html

cFE for Van Allen Probes



JHU/APL's Adoption of cFE

- **Upon completion of MESSENGER, STEREO, and New Horizons APL examined how to improve reusability of FSW**
  - Similar questions from earlier in the presentation
    - Home-grown?
    - Commercial?
    - Open Source?

- **Opted to adopt cFE from NASA/GSFC**
- **Rapid jump start in capabilities to increase reuse**
- **Still able to use heritage software by modifying to adapt to a publish and subscribe mechanism**
- **Easy addition of new applications**
- **Used on Van Allen Probes, Robotic Lunar Lander, and will be used on Solar Probe Plus**

Lessons Learned

- **Greatly reduced up-front costs for APL**
- **cFE requires mission support**
- **Centralized or community support would enable broader support and acceptance**
- **Critical time**
  - cFE is being adopted by many organizations (multiple NASA centers as well as commercial organizations)
  - However, without a method for governing development, there is a risk for disparate solutions that will end up unique at each location

Path Forward

- **Define how the community could collaborate**
- **Create a working group among interested parties to define a path of support**
- **Search for funding**
  - How to proceed?

Questions?

- **Acknowledgements: Chris Krupiarz (JHU/APL)**